

# **Beyond Monotonicity: Czech NPI Licensing Under ‘majority’**

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# Introduction

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- traditional generalization: **NPIs** are licensed under **downward-entailing operators** (Ladusaw (1979))

## 1. Some linguistic data escape this rule

(AND the weakened version by von Stechow (1999)) :

- (1)
- a. **Most** men with *any* revolutionary commitments were executed.
  - b. **Half** of those who've *ever* voted are no longer living.
  - c. **Exactly three** people did *any* work *at all*.
  - d. **Usually**, women who've *ever* seen Paris want to go back sometime.

(from Rothschild (2006a), Rothschild (2006b))

## 2. Experimental evidence remains mixed:

- Szabolcsi et al. (2008): NPIs do not facilitate downward inferences (DI)
- Chemla et al. (2011); Denić et al. (2021) : subjective judgments → DI → licensing

## 3. An alternative approach:

- Barker (2018):
  - NPIs signal **narrow scope** wrt. licensor
  - licensing = non-entailment between two scope readings
- in linguistic data, both conditions are usually met

- Dočekal and Chumchalová (2025) : double licensing structures  
*X doesn't doubt that [... NPI ...]*:
  - NPI licensing is **domain-sensitive** (Homer (2021))
  - Yet inference patterns remained **uniformly upward-entailing**
- **Problem:** DL environments still involve *monotonic* operators (negation, *doubt*) with complex scope interactions
- **Question:** Can NPIs appear in *empirically non-monotonic* environments?

## Solution:

- *Proportional* quantifier (Barwise and Cooper (1981))
- Neither downward- nor upward-entailing
- *většina* = “majority”, “most”
- $\llbracket \textit{většina} \rrbracket(A)(B) = 1$  iff  $|A \cap B| > |A \setminus B|$

- “the majority” is not DE:

(2) Context: room with 10 cats, 8 black, 2 white.

- a. The majority of cats in this room are black. True
- b. The majority of white cats in this room are black.  
Contradiction

- but it does license NPIs (variation of Rothschild's example):

(3) **The majority** of men with *any* revolutionary commitments were executed.

**Hypothesis:** NPIs are licensed under *většina* while participants simultaneously **reject both DE and UE** inferences - If true → stronger evidence against the DI–NPI link

# Experiment

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- **Platform:** PClbex Farm (Zehr and Schwarz (2018))
- Two sections: acceptability task + inference judgment task
- **Hypothesis:** PI *sebemenší* in restrictor of *most*-QP would be judged **acceptable** despite **non-monotonic** environment
- **Task:** Two sections = **acceptability** + **inference judgment**
- **Participants:** two versions based on task ordering
  - Acceptability-first: N=24
  - Inference-first: N=29
- No overlap in samples, no demographic info, option of course credits for Masaryk U. students, anonymization of datasets

## Acceptability Task

- 9 items in  $1 \times 3$  factorial design
- Conditions varied by quantifier in QP above PI:
  - DE: *all*
  - NM: *most*
  - UE: *some*
- 1–5 Likert scale
- Expectation:

*all > most > some*

### Example item:

- (4) *Všichni/Většina/Někteří občané/-ů se sebemenší vírou v Boha chtěli/-a nalezenému dítěti vystrojit křest.*

‘All/Most/Some of the citizens with even the least faith in God wanted to arrange a christening for the found child.’

- Fillers = various sentences of a similar length to the items

# Inference Task

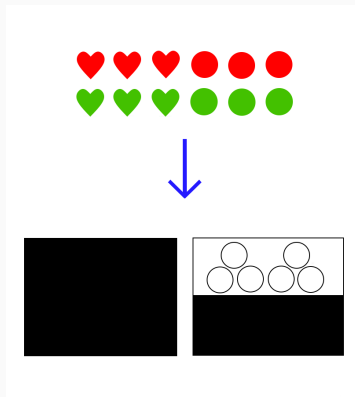
- 18 items in a  $2 \times 3$  factorial design: inference direction (UE/DE)  $\times$  quantifier (*all/most/some*)
- Participants judged whether Sentence B follows from Sentence A:

**Sentence A:** Q circles are in the box on the right.

**UE:** Q shapes are in the box on the right.

**DE:** Q green circles are in the box on the right.

*Q = all, most, or some*

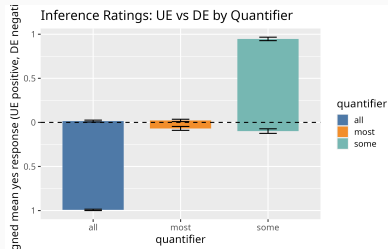


## Results

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- Acceptability ratings and inference responses shown in Figure~1
- **Analyses:** Bayesian ordinal regression (cumulative logit link) hierarchical models
- **Tool:** *R* with *brms* (Bürkner (2017))
- Version order showed no credible effect ( $|\hat{\beta}| < 0.3$ ,  $\text{BF}_{01} > 4$ )

# Acceptability Ratings & Inference Responses

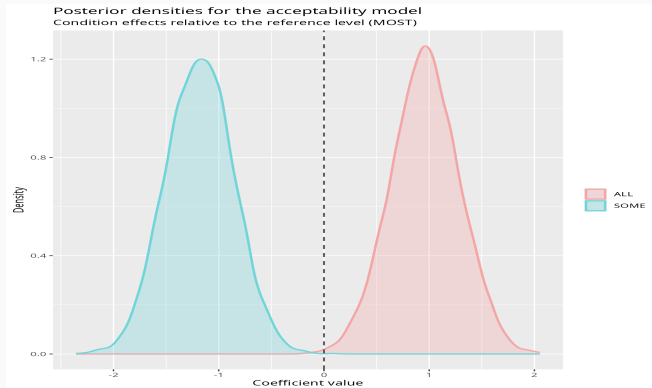


**Figure 1:** Acceptability ratings with error bars (left) and inference responses (right)

- Bayesian ordinal regression (cumulative logit link) in *R* with *brms* (Bürkner (2017))
- Weakly informative priors
- Random intercepts: by-participant and by-item (maximal models failed to converge, so simplified random effects structure used)
- **Reference level:** *most*

# Posterior Densities

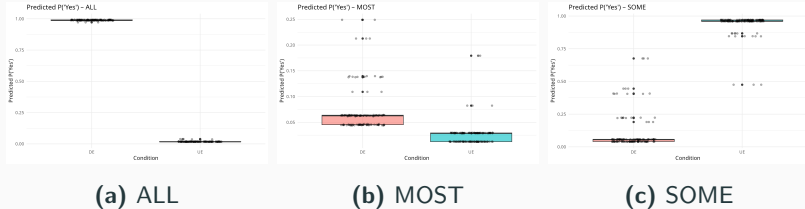
- Predicted differences on the Likert scale (the model is on the logit scale, but we report differences on the original scale, see Appendix): *all*: +0.48 points, *some*: -0.73 points



**Figure 2:** Posterior densities for the acceptability model: Condition effects relative to the reference level (MOST). ALL (red) is positively shifted, SOME (teal) is negatively shifted, both with minimal overlap at zero. <sup>14/26</sup>

## Inference Findings

- Compared inference patterns across quantifiers (*all*, *most*, *some*)
- Bayesian logistic regression (hierarchical) in *R* with *brms*
- Weakly informative priors
- Random intercepts: by-participant and by-item (maximal models failed to converge, so simplified random effects structure used)



**Figure 3:** Predicted probability distributions by quantifier.

1. *all*: DE is much higher than UE (0.963 vs 0.031), so very strong DE pattern.
2. *some*: UE is much higher than DE (0.938 vs 0.114), so very strong UE pattern.
3. *most*: both are low (0.080 and 0.025), so mostly “no” in both directions; only a weak DE tilt, not a clean DE/UE classification.

- **Dissociation confirmed:** *most* is neither downward nor upward entailing
- Despite this, *most* licenses NPIs at **intermediate levels**
- **Acceptability gradient:** *all* (+0.48 points), *most* (baseline), *some* (-0.73 points)

## Discussion

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- **Critical dissociation:** Non-monotonic *most* licenses NPIs at intermediate levels ( $\mu = 3.95$ ) despite showing **no monotonic inference pattern** (DE: 6.8%; UE: 2.3%)
- Directly challenges Ladusaw's (1979) hypothesis: **downward entailment is NOT a synchronic licensing condition** for NPIs

# Why This Matters

- Unlike DL environments (which involve monotonic operators with complex scope interactions)
- *most* is **genuinely non-monotonic** in its restrictor
- Participants robustly rejected **both** DE and UE inferences at near-floor levels (the probability of inferring DE under *most* = 0.08; UE = 0.025)
- Confirms *most* as a non-DE/non-UE environment

- Decisive monotonicity effects:
  - **Extreme** DE evidence under *all* (the probability of inferring DE under *all* = 0.963;  $BF_{10} = 3.41 \times 10^{10}$ )
  - **Extreme** UE evidence under *some* (the probability of inferring UE under *some* = 0.938;  $BF_{10} = 4.08 \times 10^{11}$ )
  - *most* yielded **no credible preference**
- Acceptability gradient mirrors monotonicity landscape:
  - Maximal DE licensing → highest acceptability
  - Genuine non-monotonicity → intermediate acceptability
  - Maximal UE → lowest acceptability

# Theoretical Implications

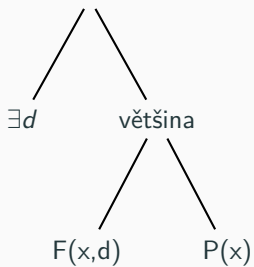
- Pattern **incompatible** with Ladusaw's synchronic DE requirement
- **Aligns** with Barker (2018)'s scope-marking view:
  - NPIs signal **narrow scope** with respect to their licensor
  - Licensed when wide scope does **not** entail narrow scope

- Formalization of the example condition:

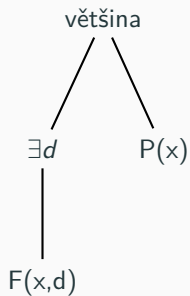
- (5) [<sub>F</sub> The majority of citizens with even the least faith in God]  
 [<sub>P</sub> wanted to arrange a christening for the found child.]
- $\exists d [\text{většina}_x(F(x, d) \rightarrow P(x))] > \text{(wide scope)}$
  - $\text{většina}_x((\exists d F(x, d)) \rightarrow P(x)) > \text{(narrow scope)}$

From  $\exists d [\text{většina}_x(F(x, d) \rightarrow P(x))]$  (there is some degree of faith such that most people satisfy  $P$ ) one cannot infer  $\text{většina}_x((\exists d F(x, d)) \rightarrow P(x))$  (most people with some degree of faith satisfy  $P$ ) since there may be a threshold of  $d$  that guarantees  $P$  (first formula true, second false)

**(5-a) Wide Scope**



**(5-b) Narrow Scope**



## Scope Entailments & NPI Licensing

Two scope readings for *sebemenší* (“even the least”):

- **Wide d:**  $\exists d [Q_x (F(x, d) \dots P(x))]$  “There is threshold *d* such that *Q* people with  $\geq d$  faith...”
- **Narrow  $\exists d$ :**  $Q_x ((\exists d F(x, d)) \dots P(x))$  “*Q* people with **any** faith...”

## Does Wide (a) entail Narrow (b)?

| Quantifier       | Entailment           | Result            | Acceptability |
|------------------|----------------------|-------------------|---------------|
| <b>ALL</b> (DE)  | NO ( $\not\models$ ) | ✓ NPI<br>Licensed | Highest       |
| <b>MOST</b> (NM) | NO ( $\not\models$ ) | ✓ NPI<br>Licensed | Intermediate  |
| <b>SOME</b> (UE) | YES ( $\models$ )    | × NOT<br>Licensed | Lowest        |

*Licensing requires **non-entailment** between scope readings*

- Under *most*, the NPI takes narrow scope within the restrictor (*sebemenší vírou*, “least faith”)
- Yields an interpretable configuration despite non-monotonicity
- Results replicate Szabolcsi et al. (2008)’s null finding
- Extends this to a theoretically critical case: an **empirically non-monotonic** quantifier
- Diachronic note: NPIs may have grammaticalized in DE environments (Herburger (2023)), but DE is **not synchronically necessary** for licensing

Thank you for your attention!



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# Appendix

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# Inference Task: Design

- 18 items in  $2 \times 3$  factorial design
- **Factor I:** we asked about upward/downward entailment
  - UE: *circle*  $\rightarrow$  *shape*
  - DE: *circle*  $\rightarrow$  *green.circle*
- **Factor II:** the quantifiers in our stimuli (*all/most/some*)
- **Visuals** – three layers of subset relations
  - **shapes:** ○ ○ ○ ○ ○ ○ ♥ ♥ ♥ ♥ ♥ ♥  
↑
  - **circles:** ○ ○ ○ ○ ○ ○  
↑
  - **green circles:** ○ ○ ○

## Inference Task: Worked Example

- BEFORE = Top of the image.
- CLEANUP = Klára places the 12 shapes into two boxes.
- AFTER = Bottom of the image.
- window in one box reveals contour but not color
- Boris knows BEFORE, did not witness CLEANUP, now sees AFTER
- AFTER = Sentence A  
Boris infers Sentence B

Sentence A:

All the *circles* are in the box on the right.

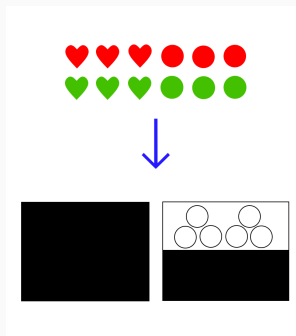
Sentence B, UE:

All the *shapes* are in the box on the right.

Sentence B, DE:

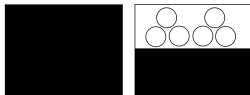
All the *green circles* are in the box on the right.

Did Boris infer B correctly? Click YES or NO!



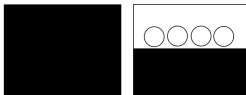
**Figure 4:** Item 14,  
conditions UE-ALL & DE-ALL

# Inference Task: Response Key



UE-ALL:  
All of the shapes?

DE-ALL:  
All of the green circles?



UE-MOST:  
Most of the shapes?

DE-MOST:  
Most of the green circles?



UE-SOME:  
Some of the shapes?

DE-SOME:  
Some of the green circles?

(Logically) correct answers:

no  
yes

no  
no

yes **Sentence A:** Q circles are in the box on the right.

**UE:** Q shapes are in the box on the right.

**DE:** Q green circles are in the box on the right.

## The acceptability part coefficients

- **Pattern:** Decisive evidence for expected ordering
  - *all* rated credibly **higher** than *most* ( $\hat{\beta} = 0.96$ , 95% CrI [0.31, 1.60],  $BF_{10} = 16.31$ : strong evidence)
  - *some* rated credibly **lower** than *most* ( $\hat{\beta} = -1.18$ , 95% CrI [-1.80, -0.55],  $BF_{10} = 104.60$ : extreme evidence)
    - *Note: coefficients on cumulative log-odds scale*

## The inference part coefficients

- **Reference:** *most* as reference level; DE as reference direction
- **Results:**
  - Extreme evidence for DE under *all* ( $\hat{\beta}_{\text{all}} = 5.71$ , 95% CrI [4.78, 6.76],  $\text{BF}_{10} = 3.41 \times 10^{10}$ )
    - plus interaction:  $\hat{\beta}_{\text{UE:all}} = -5.55$ , 95% CrI [-7.02, -4.09],  $\text{BF}_{10} = 1.04 \times 10^7$
  - Extreme evidence for UE under *some* ( $\hat{\beta}_{\text{UE:some}} = 5.99$ , 95% CrI [4.77, 7.31],  $\text{BF}_{10} = 4.08 \times 10^{11}$ )
  - **Critically:** *most* showed **no evidence** for either direction
- again coefficients on the log-odds (logit) scale, not on the probability scale

- posterior summaries on the probability scale:

| condition | median | lo95  | hi95  |
|-----------|--------|-------|-------|
| DE-most   | 0.080  | 0.039 | 0.144 |
| DE-all    | 0.963  | 0.919 | 0.985 |
| DE-some   | 0.114  | 0.061 | 0.191 |
| UE-most   | 0.025  | 0.009 | 0.062 |
| UE-all    | 0.031  | 0.010 | 0.072 |
| UE-some   | 0.938  | 0.882 | 0.973 |





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